

Screenshot showing the important Galaxy steps.

The screenshot shows the Galaxy web interface. On the left, the 'Discover 2k+ Tools' panel is visible with various bioinformatics tools. The main workspace displays a FASTA file named '5: Odocoileus virginianus_genomic.fna.gz' with 481 sequences and a size of 770.1 MB. A warning message indicates that only the first 100 KB is shown. The 'History' panel on the right shows a list of datasets, including 'Genome_Exploration_II_Bahian_Odocoileus_vir...' and '5: Odocoileus virginianus_g...'. The bottom of the interface shows a macOS dock with various application icons.

Figure 1. *Odocoileus virginianus* genome dataset and Galaxy History

The screenshot shows the Galaxy web interface with the 'gfastats' tool output for dataset 5. The main workspace displays a table with two columns: 'Column 1' and 'Column 2'. The table contains genomic statistics for dataset 5, including scaffold lengths, average scaffold lengths, and contig counts. The 'History' panel on the right shows a list of datasets, including 'Genome_Exploration_II_Bahian_Odocoileus_vir...' and '13: gfastats on dataset 5: stats'.

Column 1	Column 2
# scaffolds	481
Total scaffold length	2,423,263,331
Average scaffold length	5,037,969.59
Scaffold N50	66,396,233
Scaffold auN	65,597,579.21
Scaffold L50	15
Largest scaffold	108,574,468
Smallest scaffold	2,666
# contigs	747
Total contig length	2,423,144,861
Average contig length	3,243,835.67
Contig N50	21,664,735

Figure 2. Original genome statistics from gfastats

The screenshot shows the Galaxy web interface with the 'Compute sequence length on dataset 5' tool output. The main workspace displays a table with two columns: 'Column 1' and 'Column 2'. The table contains sequence IDs and their computed lengths. The 'History' panel on the right shows a list of datasets, including 'Genome_Exploration_II_Bahian_Odocoileus_vir...' and '7: Compute sequence length on dataset 5'.

Column 1	Column 2
NC_069674.1	198574468
NC_069675.1	192826359
NC_069676.1	196254987
NC_069677.1	93938593
NC_069678.1	93555489
NC_069679.1	89327669
NC_069680.1	85935946
NC_069681.1	86652612
NC_069682.1	78117788
NC_069683.1	73484831
NC_069684.1	71956172
NC_069685.1	68276874

Figure 3. Sequence ID and their computed lengths.

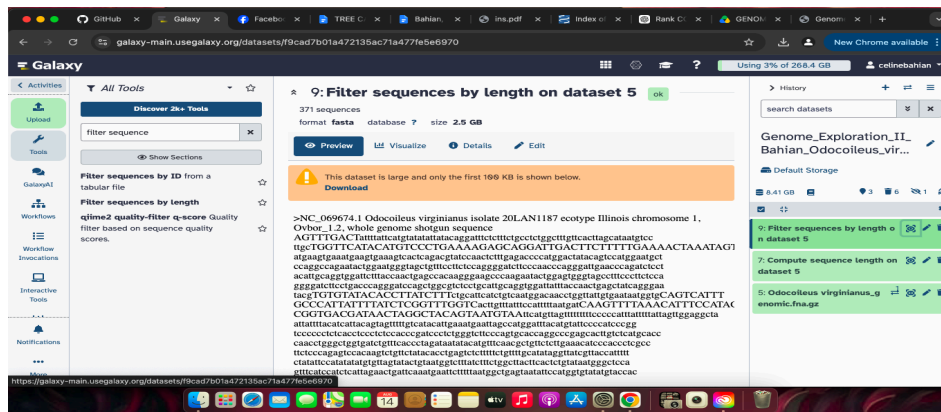


Figure 4. Result after applying the ≥ 10 kb filter.

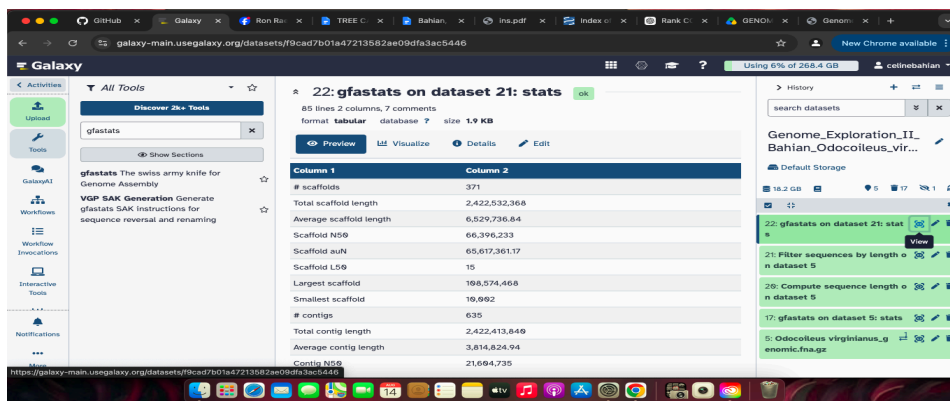


Figure 5. Statistics of the filtered assembly.

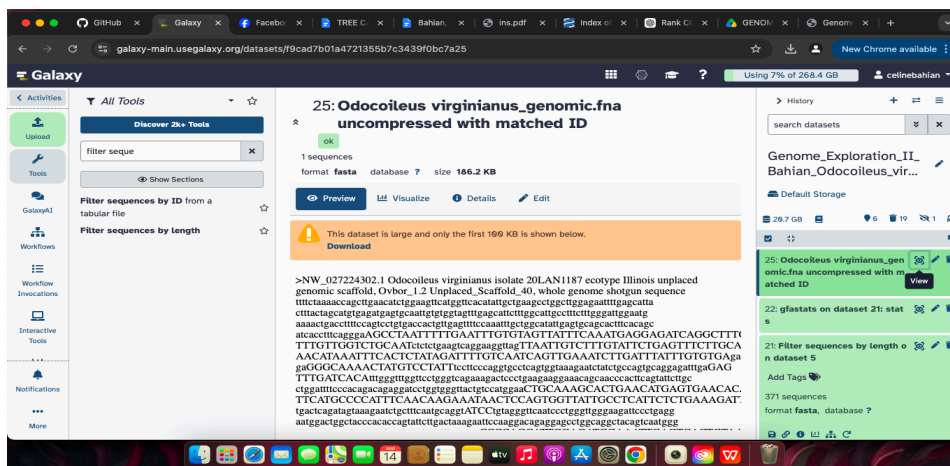


Figure 6. Selection of sequences using Filter Sequences by ID

The screenshot shows the Galaxy web interface. The main panel displays the 'getorf' tool output for dataset 21. The tool has been executed successfully, and the results are shown in a table format. The table includes the following columns: 'format', 'fasta', 'database', and 'size'. The size is 629.8 MB. The results are displayed as a list of predicted ORFs, each with its coordinates and sequence. The sequences are shown in a monospace font, with some lines wrapped. The right sidebar shows a history of datasets, including 'Genome_Exploration_II...', 'Bahian_Odocoileus_vir...', and '26: getorf on dataset 21'. The bottom of the interface shows a macOS dock with various application icons.

Figure 7. getorf output showing predicted ORF's